Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

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2026-03-10

Содержание (i)

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Цель лабораторной работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

Задачи лабораторной работы

1 Определить имя и путь домашнего каталога

2 Изучить команду ls.

3 Выполнить действия с каталогами.

4 Получить дополнительные сведения при помощи справки по командам.

5 Изучить команду history.

2. Процесс выполнения лабораторной работы

Имя и путь к домашнему каталогу

```
snbondarenko@snbondarenko:~$ cd
snbondarenko@snbondarenko:~$ pwd
/home/snbondarenko
snbondarenko@snbondarenko:~$
```

Рисунок 1: Путь к домашнему каталогу

Опции команды ls

```
snbondarenko@snbondarenko:~$ cd /tmp
snbondarenko@snbondarenko:/tmp$ ls
 11875845-276e-48de-b70e-30c29687724a.zip
 cdc6b4da-6338-473e-9449-893b80124429.zip
 happd.sock
 snap-private-tmp
 systemd-private-8b44c82296164c7db1877e277ec1c028-abrtd.service-NnKdAW
 systemd-private-8b44c82296164c7db1877e277ec1c028-chronyd.service-PcyVGi
 systemd-private-8b44c82296164c7db1877e277ec1c028-colord.service-pQN9NF
 systemd-private-8b44c82296164c7db1877e277ec1c028-dbus-broker.service-uG
0PJ8
 systemd-private-8b44c82296164c7db1877e277ec1c028-fwupd.service-WvMOV1
 systemd-private-8b44c82296164c7db1877e277ec1c028-irqbalance.service-zt4
N9v
 systemd-private-8b44c82296164c7db1877e277ec1c028-low-memory-monitor.ser
vice-L0Acr1
 systemd-private-8b44c82296164c7db1877e277ec1c028-ModemManager.service-M
65sDY
 systemd-private-8b44c82296164c7db1877e277ec1c028-pcscd.service-Gjsacu
 systemd-private-8b44c82296164c7db1877e277ec1c028-polkit.service-xHwcLG
 systemd-private-8b44c82296164c7db1877e277ec1c028-rtkit-daemon.service-p
9L22g
 systemd-private-8b44c82296164c7db1877e277ec1c028-switcheroo-control.ser
vice-jvWekq
 systemd-private-8b44c82296164c7db1877e277ec1c028-systemd-logind.service
-3DHk9d
 systemd-private-8b44c82296164c7db1877e277ec1c028-systemd-resolved.servi
ce-FExHvg
 systemd-private-8b44c82296164c7db1877e277ec1c028-upower.service-40ZcYV
'tcU0WK5PwneH++Y9c0o680J7cW+IVVsFTUXqGYhUOV8='
 tmp-y21.xpi
 vmware-root_891-3988228581
snbondarenko@snbondarenko:/tmp$
```

Рисунок 2: Команда ls

Опции команды ls

```
snbondarenko@snbondarenko:/tmp$ ls -a
 .
 ..
 11875845-276e-48de-b70e-30c29687724a.zip
 cdc6b4da-6338-473e-9449-893b80124429.zip
 .font-unix
 happd.sock
 .ICE-unix
 snap-private-tmp
 systemd-private-8b44c82296164c7db1877e277ec1c028-abrtd.service-NnKdAW
 systemd-private-8b44c82296164c7db1877e277ec1c028-chronyd.service-PcyVGi
 systemd-private-8b44c82296164c7db1877e277ec1c028-colord.service-pQN9NF
 systemd-private-8b44c82296164c7db1877e277ec1c028-dbus-broker.service-uG
0PJB
 systemd-private-8b44c82296164c7db1877e277ec1c028-fwupd.service-WvMOV1
 systemd-private-8b44c82296164c7db1877e277ec1c028-irqbalance.service-zt4
N9v
 systemd-private-8b44c82296164c7db1877e277ec1c028-low-memory-monitor.ser
vice-L0Acr1
 systemd-private-8b44c82296164c7db1877e277ec1c028-ModemManager.service-M
G5sDY
 systemd-private-8b44c82296164c7db1877e277ec1c028-pcscd.service-Gjsacu
 systemd-private-8b44c82296164c7db1877e277ec1c028-polkit.service-xHwcLG
 systemd-private-8b44c82296164c7db1877e277ec1c028-rtkit-daemon.service-p
9L22g
 systemd-private-8b44c82296164c7db1877e277ec1c028-switcheroo-control.ser
vice-jvWekq
 systemd-private-8b44c82296164c7db1877e277ec1c028-systemd-logind.service
-3DHk9d
 systemd-private-8b44c82296164c7db1877e277ec1c028-systemd-resolved.servi
ce-FExHvg
 systemd-private-8b44c82296164c7db1877e277ec1c028-upower.service-40ZcYV
'tcU0WK5PwneH++Y9c0o680J7cW+IVVsFTUXqGYhUOV8='
 tmp-y21.xpi
 vmware-root_891-3988228581
 .X0-lock
```

Рисунок 3: Команда ls -a

Опции команды ls

```
snbondarenko@snbondarenko:/tmp$ ls -l
total 9696
-rw-r--r--. 1 snbondarenko snbondarenko 4182728 Mar 10 15:04  11875845-2
76e-48de-b70e-30c29887724a.zip
-rw-r--r--. 1 snbondarenko snbondarenko 4182728 Mar 10 15:04  cdc6b4da-6
338-473e-9449-893b80124429.zip
srw-rw-rw-. 1 root         root                0 Mar 10 14:59  happd.sock
drwx------. 2 root         root               40 Mar 10 14:58  snap-priva
te-tmp
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-abrtd.service-NnKdAW
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-chronyd.service-PcyVGi
drwx------. 3 root         root               60 Mar 10 14:59  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-colord.service-pQN9NF
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-dbus-broker.service-uG0PJ8
drwx------. 3 root         root               60 Mar 10 14:59  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-fwupd.service-WvMOV1
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-irqbalance.service-zt4N9v
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-low-memory-monitor.service-L0Acr1
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-ModemManager.service-M65sDY
drwx------. 3 root         root               60 Mar 10 14:59  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-pcscd.service-Gjsacu
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-polkit.service-xHwcLG
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-rtkit-daemon.service-p9L22g
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
ivate-8b44c82296164c7db1877e277ec1c028-switcheroo-control.service-jvWekq
drwx------. 3 root         root               60 Mar 10 14:58  systemd-pr
```

Рисунок 4: Команда ls -l

Опции команды ls

```
snbondarenko@snbondarenko:/tmp$ ls -f
 .
 ..
'tcU0WK5PwneH++Y9c0o680J7cW+IVVsFTUXqGYhUOV8='
 cdc6b4da-6338-473e-9449-893b80124429.zip
 11875845-276e-48de-b70e-30c29687724a.zip
 tmp-y21.xpi
 systemd-private-8b44c82296164c7db1877e277ec1c028-fwupd.service-WvMOV1
 .X1-lock
 .X0-lock
 systemd-private-8b44c82296164c7db1877e277ec1c028-pcscd.service-Gjsacu
 systemd-private-8b44c82296164c7db1877e277ec1c028-colord.service-pQN9NF
 happd.sock
 systemd-private-8b44c82296164c7db1877e277ec1c028-ModemManager.service-M
65sDY
 vmware-root_891-3988228581
 systemd-private-8b44c82296164c7db1877e277ec1c028-systemd-logind.service
-3DHk9d
 systemd-private-8b44c82296164c7db1877e277ec1c028-upower.service-40ZcYV
 systemd-private-8b44c82296164c7db1877e277ec1c028-switcheroo-control.ser
vice-jvWekq
 systemd-private-8b44c82296164c7db1877e277ec1c028-rtkit-daemon.service-p
9L22g
 systemd-private-8b44c82296164c7db1877e277ec1c028-polkit.service-xHwcLG
 systemd-private-8b44c82296164c7db1877e277ec1c028-low-memory-monitor.ser
vice-L0Acr1
 systemd-private-8b44c82296164c7db1877e277ec1c028-chronyd.service-PcyVGi
 systemd-private-8b44c82296164c7db1877e277ec1c028-irqbalance.service-zt4
N9v
 systemd-private-8b44c82296164c7db1877e277ec1c028-abrtd.service-NnKdAW
 systemd-private-8b44c82296164c7db1877e277ec1c028-dbus-broker.service-uG
0PJ8
```

Рисунок 5: Команда ls -f

Каталог /var/spool

```
snbondarenko@snbondarenko:/tmp$
snbondarenko@snbondarenko:/tmp$ cd /var/spool/
snbondarenko@snbondarenko:/var/spool$ ls -l
total 0
drwxr-x--x. 1 root abrt 11924 Mar 10 14:59 abrt
drwx------. 1 abrt abrt     0 Dec  4 03:00 abrt-upload
drwxr-xr-x. 1 root root    66 Oct 23 07:05 anacron
drwx------. 1 root root    18 Oct 23 07:07 at
drwx------. 1 root root     0 Aug  6  2025 cron
drwx--x---. 1 root lp       6 Dec 12 03:00 cups
drwxr-xr-x. 1 root root     0 Jul 30  2025 lpd
drwxrwxr-x. 1 root mail  1466 Mar  7 12:48 mail
drwxr-xr-x. 1 root root     0 Jan 27 03:00 plymouth
snbondarenko@snbondarenko:/var/spool$
```

Рисунок 6: Каталог /var/spool

Домашний каталог

```
snbondarenko@snbondarenko:/var/spool$ cd
snbondarenko@snbondarenko:~$ ls
Desktop    Downloads    Music      Public   snap        Videos
Documents  git-extended Pictures   site     Templates   work
snbondarenko@snbondarenko:~$ ls -al
total 24
drwx------. 1 snbondarenko snbondarenko  382 Mar  3 15:14 .
drwxr-xr-x. 1 root         root         1466 Mar  7 12:48 ..
-rw-------. 1 snbondarenko snbondarenko 5241 Mar  3 15:34 .bash_history
-rw-r--r--. 1 snbondarenko snbondarenko   18 Jul 23  2025 .bash_logout
-rw-r--r--. 1 snbondarenko snbondarenko  144 Jul 23  2025 .bash_profile
-rw-r--r--. 1 snbondarenko snbondarenko  682 Mar  3 15:07 .bashrc
drwx------. 1 snbondarenko snbondarenko  492 Mar  3 15:13 .cache
drwx------. 1 snbondarenko snbondarenko  364 Mar 10 15:10 .config
drwxr-xr-x. 1 snbondarenko snbondarenko    0 Mar  3 14:52 Desktop
drwxr-xr-x. 1 snbondarenko snbondarenko    0 Mar  3 14:52 Documents
drwxr-xr-x. 1 snbondarenko snbondarenko    0 Mar  3 14:52 Downloads
-rw-r--r--. 1 snbondarenko snbondarenko  238 Mar  3 14:56 .gitconfig
drwxr-xr-x. 1 snbondarenko snbondarenko   74 Mar  3 15:10 git-extended
drwx------. 1 snbondarenko snbondarenko  136 Mar  3 15:34 .gnupg
drwx------. 1 snbondarenko snbondarenko   20 Mar  3 14:52 .local
drwxr-xr-x. 1 snbondarenko snbondarenko   34 Oct 23 07:05 .mozilla
drwxr-xr-x. 1 snbondarenko snbondarenko    0 Mar  3 14:52 Music
drwxr-xr-x. 1 snbondarenko snbondarenko   26 Mar  3 15:14 .npm
drwxr-xr-x. 1 snbondarenko snbondarenko    0 Mar  3 14:52 Pictures
drwxr-xr-x. 1 snbondarenko snbondarenko    0 Mar  3 14:52 Public
drwxr-xr-x. 1 snbondarenko snbondarenko   58 Mar  3 15:14 site
drwx------. 1 snbondarenko snbondarenko    8 Mar  3 15:14 snap
drwx------. 1 snbondarenko snbondarenko  132 Mar  3 14:57 .ssh
drwxr-xr-x. 1 snbondarenko snbondarenko    0 Mar  3 14:52 Templates
drwxr-xr-x. 1 snbondarenko snbondarenko   18 Mar  3 15:02 .texlive2025
drwxr-xr-x. 1 snbondarenko snbondarenko    0 Mar  3 14:52 Videos
drwxr-xr-x. 1 snbondarenko snbondarenko   10 Mar  3 14:57 work
snbondarenko@snbondarenko:~$
```

Рисунок 7: Файлы в домашнем каталоге

Работа с каталогами

```
snbondarenko@snbondarenko:~$ mkdir newdir
snbondarenko@snbondarenko:~$ mkdir newdir/morefun
snbondarenko@snbondarenko:~$ mkdir letters memos misk
snbondarenko@snbondarenko:~$ ls
Desktop    git-extended  misk    Pictures  snap       work
Documents  letters       Music   Public    Templates
Downloads  memos         newdir  site      Videos
snbondarenko@snbondarenko:~$ rm letters/ memos/ misk/
rm: cannot remove 'letters/': Is a directory
rm: cannot remove 'memos/': Is a directory
rm: cannot remove 'misk/': Is a directory
snbondarenko@snbondarenko:~$ rm -r letters/ memos/ misk/
snbondarenko@snbondarenko:~$ rm -r newdir/
snbondarenko@snbondarenko:~$ ls
Desktop    Downloads     Music     Public  snap       Videos
Documents  git-extended  Pictures  site    Templates  work
snbondarenko@snbondarenko:~$
```

Рисунок 8: Действия с каталогами

Опции команды ls

```
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/tem
plate/report/report/image':
solvay.jpg

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/tem
plate/report/report/_resources':
csl  tex

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/tem
plate/report/report/_resources/csl':
gost-r-7-0-5-2008-numeric.csl

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/tem
plate/report/report/_resources/tex':
preamble.tex

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/tem
plate/report/scripts':
image-report  mpv-shot
snbondarenko@snbondarenko:~$ ls -t
snap  git-extended  Desktop    Downloads  Pictures  Templates
site  work          Documents  Music      Public    Videos
snbondarenko@snbondarenko:~$
```

Рисунок 9: Команда ls -R и ls -t

Справка по командам

```
BASH_BUILTINS(1)                    General Commands Manual                    BASH_BUILTINS(1)

NAME
       :, ., [, alias, bg, bind, break, builtin, caller, cd, command, compgen, complete, compopt,
       continue,  declare,  dirs,  disown, echo, enable, eval, exec, exit, export, false, fc, fg,
       getopts, hash, help, history, jobs, kill,  let,  local,  logout,  mapfile,  popd,  printf,
       pushd,  pwd,  read, readarray, readonly, return, set, shift, shopt, source, suspend, test,
       times, trap, true, type, typeset, ulimit, umask, unalias, unset, wait - bash built-in com-
       mands, see bash(1)

BASH BUILTIN COMMANDS
       Unless otherwise noted, each builtin command documented in this section as  accepting  op-
       tions preceded by - accepts -- to signify the end of the options.  The :, true, false, and
       test/[  builtins  do  not accept options and do not treat -- specially.  The exit, logout,
       return, break, continue, let, and shift builtins accept and  process  arguments  beginning
       with  -  without requiring --.  Other builtins that accept arguments but are not specified
       as accepting options interpret arguments beginning with - as invalid options  and  require
       -- to prevent this interpretation.

       : [arguments]
              No  effect;  the command does nothing beyond expanding arguments and performing any
              specified redirections.  The return status is zero.

       . [-p path] filename [arguments]
       source [-p path] filename [arguments]
              The . command (source) reads and execute commands  from  filename  in  the  current
              shell  environment  and  returns  the exit status of the last command executed from
              filename.
 Manual page cd(1) line 1 (press h for help or q to quit)
```

Рисунок 10: Справка по команде cd

Справка по командам

```
PWD(1)                                User Commands                                PWD(1)

NAME
       pwd - print name of current/working directory

SYNOPSIS
       pwd [OPTION]...

DESCRIPTION
       Print the full filename of the current working directory.

       -L, --logical
              use PWD from environment, even if it contains symlinks

       -P, --physical
              resolve all symlinks

       --help display this help and exit

       --version
              output version information and exit

       If no option is specified, -P is assumed.

       Your shell may have its own version of pwd, which usually supersedes the version described
       here.   Please  refer  to your shell's documentation for details about the options it sup-
       ports.

 Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

Справка по командам

```
MKDIR(1)                          User Commands                          MKDIR(1)

NAME
       mkdir - make directories

SYNOPSIS
       mkdir [OPTION]... DIRECTORY...

DESCRIPTION
       Create the DIRECTORY(ies), if they do not already exist.

       Mandatory arguments to long options are mandatory for short options too.

       -m, --mode=MODE
              set file mode (as in chmod), not a=rwx - umask

       -p, --parents
              no  error if existing, make parent directories as needed, with their file modes un-
              affected by any -m option

       -v, --verbose
              print a message for each created directory

       -Z     set SELinux security context of each created directory to the default type

       --context[=CTX]
              like -Z, or if CTX is specified then set the SELinux or SMACK security  context  to
              CTX
 Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

Справка по командам

```
RMDIR(1)                          User Commands                          RMDIR(1)

NAME
       rmdir - remove empty directories

SYNOPSIS
       rmdir [OPTION]... DIRECTORY...

DESCRIPTION
       Remove the DIRECTORY(ies), if they are empty.

       --ignore-fail-on-non-empty
              ignore each failure to remove a non-empty directory

       -p, --parents
              remove  DIRECTORY  and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b
              a'

       -v, --verbose
              output a diagnostic for every directory processed

       --help display this help and exit

       --version
              output version information and exit

AUTHOR
       Written by David MacKenzie.
 Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

Справка по командам

```
RM(1)                                User Commands                                RM(1)

NAME
       rm - remove files or directories

SYNOPSIS
       rm [OPTION]... [FILE]...

DESCRIPTION
       This manual page documents the GNU version of rm.  rm removes each specified file.  By de-
       fault, it does not remove directories.

       If  the  -I  or --interactive=once option is given, and there are more than three files or
       the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed with
       the entire operation.  If the response is not affirmative, the entire command is aborted.

       Otherwise, if a file is unwritable, standard input is a terminal, and the  -f  or  --force
       option  is  not  given,  or the -i or --interactive=always option is given, rm prompts the
       user for whether to remove the file.  If the response is  not  affirmative,  the  file  is
       skipped.

OPTIONS
       Remove (unlink) the FILE(s).

       -f, --force
              ignore nonexistent files and arguments, never prompt

       -i     prompt before every removal
 Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

История командной строки

```
276  cd /tmp
277  ls
278  ls -a
279  ls -l
280  ls -f
281  cd /var/spool/
282  ls -l
283  cd
284  ls
285  ls -al
286  mkdir newdir
287  mkdir newdir/morefun
288  mkdir letters memos misk
289  ls
290  rm letters/ memos/ misk/
291  rm -r letters/ memos/ misk/
292  rm -r newdir/
293  ls
294  ls -R
295  ls -t
296  help cd
297  man pwd
298  man mkdir
```

Рисунок 15: Команда history

3. Выводы по проделанной работе

Вывод

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.